

Andreas Tzitzikas

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Education

University of Maryland, College Park, MD Expected May 2027
Master of Science in Computer Engineering **GPA: 4.00**
Bachelor of Science in Computer Engineering (GPA: 3.88) Expected May 2026
Relevant Coursework: Electric Circuits (ENEE205), Elements of Discrete Signal Analysis (ENEE222), Digital Circuits Lab (ENEE245), Introduction to Micro/Nanoelectronics (ENEE304), Microprocessors (ENEE440)

Projects

STM32G4 Bare-Metal Peripheral Integration | ARM Assembly, Analog/Digital Interfaces Jan 2025 – May 2025

- Implemented direct register-level programming for ADC, DAC, PWM/PFM generation using TIM2, and SPI flash memory interfaces on STM32G4 microcontroller platform.
- Engineered multiplexed interrupt system routing single hardware interrupt to multiple peripheral drivers with state-based arbitration for efficient resource utilization.
- Developed comprehensive analog signal processing including ADC sampling, DAC output generation, and PWM motor control with precise timing and synchronization requirements.

Dual-Host S2-LP Radio Shield | KiCad, RF Design, Multi-Platform Firmware Jan 2024 – Present

- Designed dual-host development board for S2-LP sub-1 GHz transceiver with STM32 and RP2350 compatibility, implementing complete RF circuit with matching network and crystal oscillator for optimal radio performance.
- Developed shadow mode debugging architecture where STM32 passively monitors SPI traffic between RP2350 and radio, enabling cross-platform firmware migration and protocol analysis for wireless development.
- Created comprehensive firmware suite for both platforms including SPI drivers, radio configuration, packet transmission, and bus monitoring capabilities with detailed technical documentation.

Tomasulo Out-of-Order RISC-V CPU | Digital Design, Timing Analysis, Verification Nov 2025 - Dec 2025

- Architected out-of-order execution system using Tomasulo's Algorithm with register renaming, reservation stations, and reorder buffer for advanced digital system design.
- Achieved 36% higher operating frequency (69.7MHz vs 51.2MHz) and 40-150% IPC improvement compared to in-order implementation through timing optimization and performance analysis.
- Synthesized to fabrication-ready GDSII using Sky130 130nm PDK, achieving zero DRC/LVS errors and comprehensive static timing analysis for reliable digital circuit operation.

Experience

Embedded Systems Intern, Alchemy – College Park, MD Jun 2025 – Aug 2025

- Engineered real-time control systems for hybrid solid oxide fuel cell reactors, implementing precise analog signal processing and motor control with PWM and ADC interfaces.
- Developed embedded firmware with interrupt-driven architectures, real-time scheduling, and robust error handling for industrial control applications.
- Integrated analog sensors and actuators with digital control systems, ensuring signal integrity, noise immunity, and reliable operation in harsh environments.

Undergraduate Teaching Assistant, ENEE205 Electric Circuits – College Park, MD Sep 2025 – Present

- Lead laboratory and discussion sections covering circuit theory fundamentals, Kirchhoff's laws, Thevenin/Norton equivalents, and AC/DC circuit analysis.
- Provide hands-on guidance with oscilloscopes, signal generators, and circuit simulation tools for practical circuit design and measurement techniques.

Technical Coordinator, Electronics Prototyping Lab, Terrapin Works – College Park, MD Jan 2024 – Present

- Maintain and operate LPKF PCB prototyping equipment ensuring reliable fabrication capabilities with 90% uptime for student and faculty projects.
- Provide technical support for PCB design, fabrication, and assembly processes including schematic capture, layout, and component soldering techniques.

Skills

Circuit Design: Analog/Digital Circuits, Signal Processing, Power Systems, Filter Design
Electronics: Microcontrollers, ADC/DAC, PWM, SPI/I2C/UART, GPIO, Embedded Programming
Laboratory Tools: Oscilloscopes, Signal Generators, Multimeters, PCB Fabrication
Software Tools: KiCad, Cadence Spectre, MATLAB, LTspice, Xilinx Vivado
Analysis: Circuit Analysis, Signal Processing, Control Systems, Electromagnetic Compatibility